SIMATS ENGINEERING
ASSESSMENT TOOL 3(Simulation Task)

COURSE NAME: Control Systems
COURSE CODE: EEA05

10. Compare frequency response of first-order of a given transfer function G(s) = and second-order systems of a given transfer function G(s) = through simulation and analyze differences.

Program:

```
clc;
clear;
clf;

// ----------------------------------------
// Define Laplace variable
// ----------------------------------------
s = poly(0, "s");

// ----------------------------------------
// Transfer functions
// ----------------------------------------
G1 = 1/(s + 1);               // First-order
G2 = 25/(s^2 + 4*s + 25);      // Second-order
```

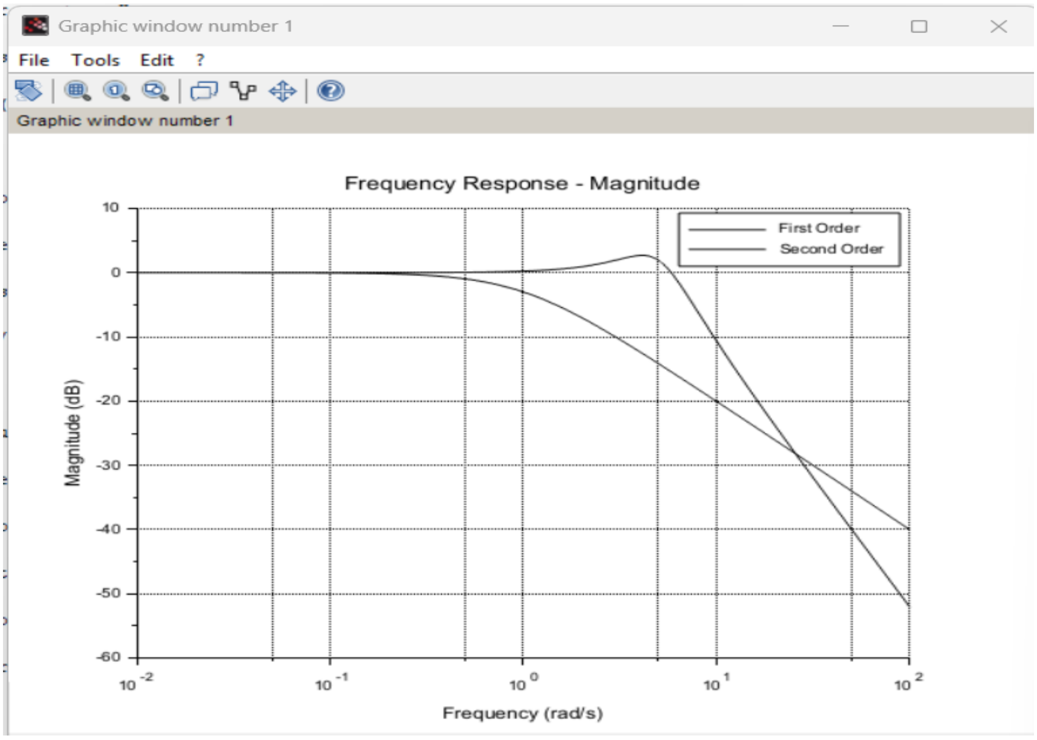
```
// ----------------------------------------
// Frequency range
// ----------------------------------------
w = logspace(-2, 2, 500);

// ----------------------------------------
// Frequency response using horner()
// ----------------------------------------
H1 = horner(G1, %i*w);
H2 = horner(G2, %i*w);

// ----------------------------------------
// MAGNITUDE
// ----------------------------------------
Mag1 = abs(H1);
Mag2 = abs(H2);

// Convert magnitude to dB
Mag1_dB = 20*log10(Mag1);
Mag2_dB = 20*log10(Mag2);

// ----------------------------------------
// PHASE
// ----------------------------------------
Phase1 = atan(imag(H1), real(H1)) * 180/%pi;
Phase2 = atan(imag(H2), real(H2)) * 180/%pi;


//
========================================
```

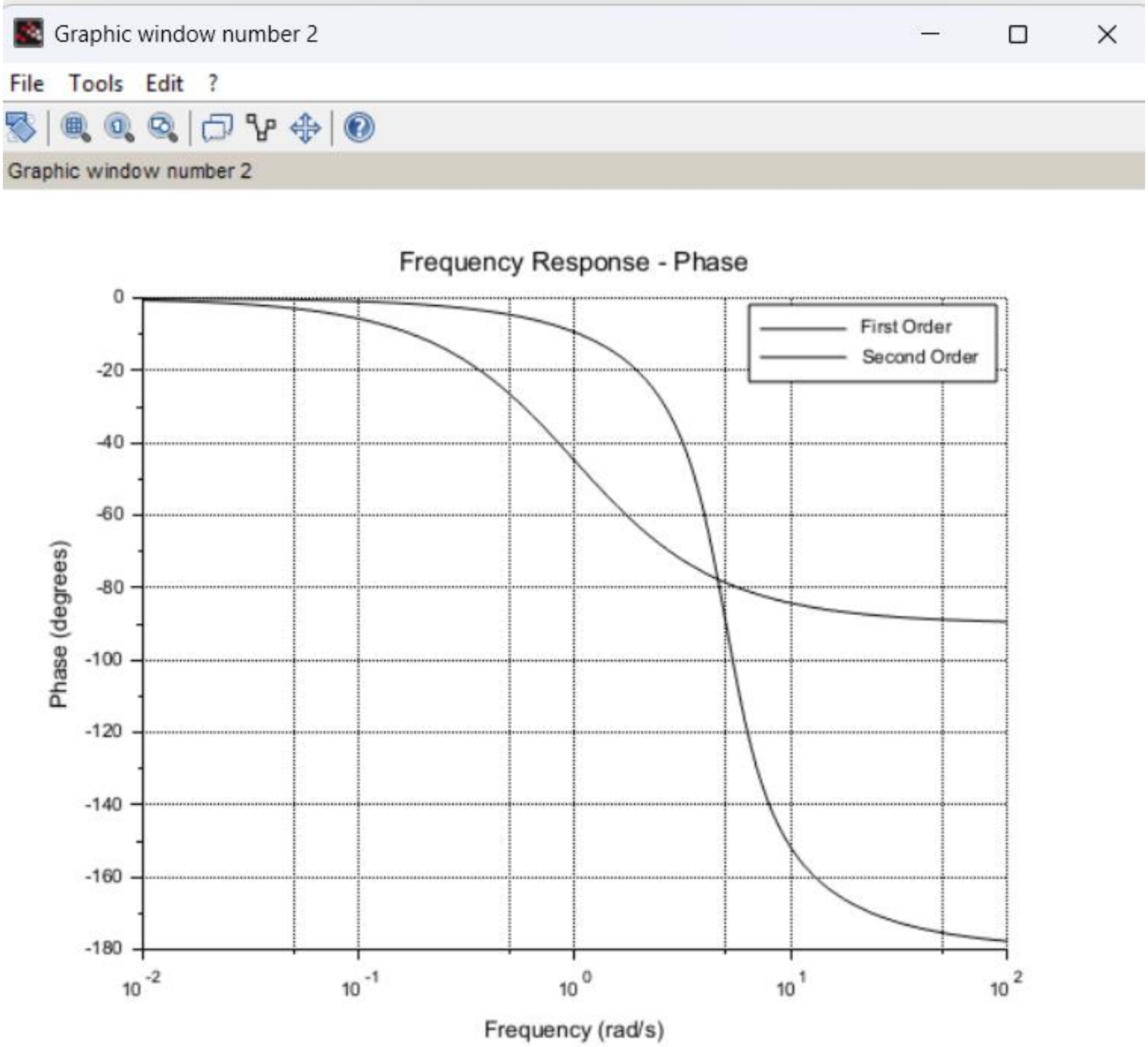
```
==
// FIGURE 1 : MAGNITUDE RESPONSE
//
========================================
==
scf(1);
clf();

plot2d("ln", w, Mag1_dB);
plot2d("ln", w, Mag2_dB);

xgrid();

xlabel("Frequency (rad/s)");
ylabel("Magnitude (dB)");
title("Frequency Response - Magnitude");

legend(["First Order"; "Second Order"],
"in_upper_right");


//
========================================
==
// FIGURE 2 : PHASE RESPONSE
//
========================================
==
scf(2);
clf();
```

```
plot2d("ln", w, Phase1);
plot2d("ln", w, Phase2);

xgrid();

xlabel("Frequency (rad/s)");
ylabel("Phase (degrees)");
title("Frequency Response - Phase");

legend(["First Order"; "Second Order"],
"in_upper_right");


//
==========================================
==
// DISPLAY IMPORTANT VALUES
//
==========================================
==
disp("First-order system:");
disp("G1(s) = 1/(s+1)");

disp("Second-order system:");
disp("G2(s) = 25/(s^2+4s+25)");

disp("Natural frequency of second-order system = 5
rad/s");
disp("Damping ratio of second-order system = 0.4");
```

```
disp("Resonant frequency = 4.123 rad/s");
```

OUTPUT:





Graphic window number 2
File Tools Edit ?
Graphic window number 2
Frequency Response - Phase
First Order
Second Order
Phase (degrees)
0
-20
-40
-60
-80
-100
-120
-140
-160
-180
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
Frequency (rad/s)